PROBLEM STATEMENT ONE: MILLION TOKEN ENGINEERING · MANAGED AGENTS

Argon Speed

AI that makes itself faster. Gemini 4 Argon, running on Google's Managed Agents, rewrites the GPU kernels that AI models run on, around the clock. Today: Muse Spark Open's. Next: its own.

816B

parameters in Muse Spark Open, served by SGLang on 8×H200

5

GPU kernels that decide its speed, from projections to a whole decoder layer

2.5–11×

the gap between SGLang's production kernels and the speed of light

24/7 ON GOOGLE'S MANAGED AGENTS

A research loop with an independent judge

Google trigger
wakes each problem's agent every hour

→

Gemini 4 Argon
in the problem's own persistent sandbox

→

okc check
the judge times it on an H200 against SGLang

→

NOTES.md
what it learned and the next step

No secrets in the sandbox

20 checks a day, budgeted

Compiles before it checks

People approve scored runs

MILLION-TOKEN ENGINEERING

It reads the whole stack, then writes the kernel

58M

tokens read, 94% from cache: SGLang's source at the judge's commit, its environment, per-shape profiles

742

lines of CUDA kernel from a team design spec, written in one session

9 / 9

hidden correctness tests passed on its first check

RESULTS, AS THE JUDGE MEASURED THEM

Faster than SGLang on all 5 problems

PROBLEM	HEADROOM	OUR BEST CHECK	SCORED / BOARD	BOARD LEADER
Dense projections BF16 projections and the FP32 router	2.5×	**1.092×**	#921 of 1,007	1.61×
Attention decode gated GQA over a paged FP8 cache	11.4×	**1.260×**	#693 of 742	2.91×
Routed experts top-8 of 256 experts, NVFP4 weights	4.4×	**1.323×**	#568 of 791	2.55×
Decoder layer a whole layer at decode shapes	3.9×	**1.226×**	#534 of 690	2.31×
Prefill layer a whole layer at prefill shapes	3.0×	**1.062×**	#368 of 425	1.96×

#597 of 1,149 teams on the overall board. $18.12 spent so far; about $134 a day, estimated, for all five 24/7.

THE SPEED IN ARGON SPEED

Aimed at its own serving, it compounds

Faster kernels

↓

Cheaper, quicker agent runs

↓

More iterations a day

so the next speedup comes sooner

14%

off every decode step at our own decode-layer kernel (1.23×)

$86M

a year on 50,000 GPUs, and 12,787 years of user waiting saved (illustrative)